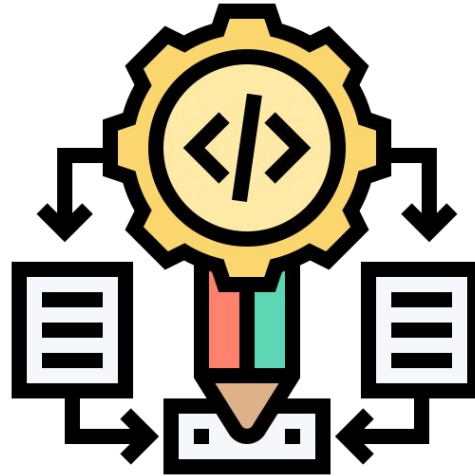


Algorithm 1



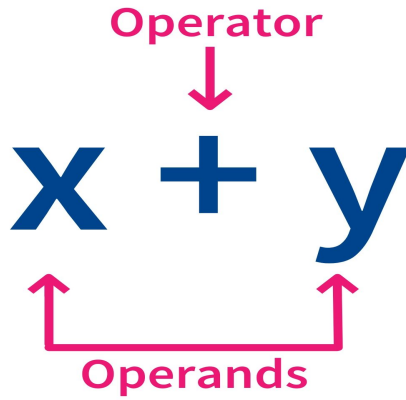
Dr Benabderrezak
Researcher in Cyber Security
Full-stack Web developer

Operators

العوامل



Operators



Operators

1. **Arithmetic** operators العوامل الحسابية
2. **Assignment** operators عوامل الإسناد
3. **Comparison** operators عوامل المقارنة
4. **Logical** operators العوامل المنطقية

1. Arithmetic operators

Operator	Name	Description	Example
+	Addition	Adds together two values	$x + y$
-	Subtraction	Subtracts one value from another	$x - y$
*	Multiplication	Multiplies two values	$x * y$
/	Division	Divides one value by another	x / y
%	Modulus	Returns the division remainder	$x \% y$
++	Increment	Increases the value of a variable by 1	$++x$
--	Decrement	Decreases the value of a variable by 1	$--x$

1. Arithmetic operators

```
#include <stdio.h>
void main() {
    int a = 10, b = 5;

    printf("Addition: %d\n", a+b);
    printf("Subtraction: %d\n", a-b);
    printf("Multiplication: %d\n", a*b);
    printf("Division: %d\n", a/b);
    printf("Modulus: %d\n", a%b);

    printf("Pre-Increment: %d\n", ++a);
    printf("Post-Increment: %d\n", a++);

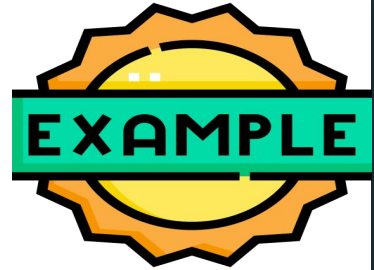
    printf("Pre-Decrement: %d\n", --a);
    printf("Post-Decrement: %d\n", --a);
}
```



```
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2
Modulus: 0
Pre-Increment: 11
Post-Increment: 11
Pre-Decrement: 11
Post-Decrement: 10
```

2. Assignment operators

Operator	Example	Same As
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
%=	x %= 3	x = x % 3



```
#include <stdio.h>
void main() {
    int num1 = 10, num2 = 5, result;

    result = num1;
    printf("Initial value of result: %d\n", result);

    result += num2;
    printf("After addition assignment: %d\n", result);

    result -= num2;
    printf("After subtraction assignment: %d\n", result);
}
```



```
Initial value of result: 10
After addition assignment: 15
After subtraction assignment: 10
```

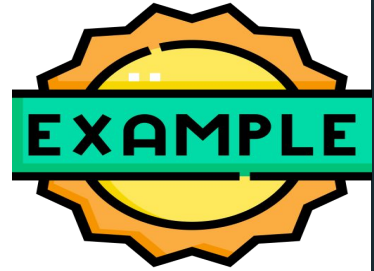
```
result *= num2;  
printf("After multiplication assignment: %d\n", result);  
  
result /= num2;  
printf("After division assignment: %d\n", result);  
  
result %= num2;  
printf("After modulus assignment: %d\n", result);  
  
}
```



```
After multiplication assignment: 50  
After division assignment: 10  
After modulus assignment: 0
```

3. Comparison operators

Operator	Name	Example	Description
==	Equal to	<code>x == y</code>	Returns 1 if the values are equal
!=	Not equal	<code>x != y</code>	Returns 1 if the values are not equal
>	Greater than	<code>x > y</code>	Returns 1 if the first value is greater than the second value
<	Less than	<code>x < y</code>	Returns 1 if the first value is less than the second value
>=	Greater than or equal to	<code>x >= y</code>	Returns 1 if the first value is greater than, or equal to, the second value
<=	Less than or equal to	<code>x <= y</code>	Returns 1 if the first value is less than, or equal to, the second value



```
#include <stdio.h>
```

```
void main() {
```

```
int num1 = 10;
```

```
int num2 = 5;
```

```
// Equal to comparison
```

```
printf("%d is equal to %d : %d\n", num1, num2, num1==num2);
```

```
// Not equal to comparison
```

```
printf("%d is not equal to %d : %d\n", num1, num2, num1 != num2);
```



```
10 is equal to 5 : 0
```

```
10 is not equal to 5 : 1
```

```
// Greater than comparison
```

```
printf("%d is greater than %d : %d\n", num1, num2, num1 > num2 );
```

```
// Less than comparison
```

```
printf("%d is less than %d : %d\n", num1, num2 , num1 < num2);
```



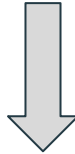
```
10 is greater than 5 : 1  
10 is less than 5 : 0
```

```
// Greater than or equal to comparison
```

```
printf("%d is greater than or equal to %d : %d \n", num1, num2 , num1 >= num2);
```

```
// Less than or equal to comparison
```

```
printf("%d is less than or equal to %d : %d\n", num1, num2 , num1 <= num2);
```



```
10 is greater than or equal to 5 : 1  
10 is less than or equal to 5 : 0
```

3. Logical operators

Operator	Name	Example	Description
&&	Logical and	<code>x < 5 && x < 10</code>	Returns 1 if both statements are true
	Logical or	<code>x < 5 x < 4</code>	Returns 1 if one of the statements is true
!	Logical not	<code>!(x < 5 && x < 10)</code>	Reverse the result, returns 0 if the result is 1

```
#include <stdio.h>
void main() {
int num1 = 10,num2 = 5,num3 = 15;
// AND operator
if (num1 > num2 && num1 < num3) {
    printf("%d is greater than %d and less than %d\n", num1, num2, num3);
} else {
    printf("%d is not greater than %d and less than %d\n", num1, num2, num3);
}
```



10 is greater than 5 and less than 15

```
#include <stdio.h>
void main() {
int num1 = 10,num2 = 5,num3 = 15;
// OR operator
if (num1 > num2 || num1 > num3) {
    printf("%d is greater than %d or %d\n", num1, num2, num3);
} else {
    printf("%d is not greater than %d or %d\n", num1, num2, num3);
}
```



10 is greater than 5 or 15

```
#include <stdio.h>
void main() {
    int num1 = 10, num2 = 5, num3 = 15;

    // NOT operator
    if (!(num1 == num2)) {
        printf("%d is not equal to %d\n", num1, num2);
    } else {
        printf("%d is equal to %d\n", num1, num2);
    }
}
```



10 is not equal to 5



Thank
You!

